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The Editors
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Dear Editors,

We are pleased to submit our manuscript, **"Scaffold-Controlled Evaluation of Molecular Representations for Antimalarial Activity Prediction"**, for consideration as a Research Article in the *Journal of Computer-Aided Molecular Design*.

The manuscript presents a scaffold-controlled comparison of graph, sequence, and topological molecular representations against an ECFP4–random-forest reference. We test this question on a frozen panel of 19 836 African antimalarial natural products under random and Bemis–Murcko scaffold cross-validation (25 fold–seed replicates), comparing GIN, GIN–TFP, GIN–TNE, and ChemBERTa with ECFP4–RF.

Two contributions matter for readers of JCAMD. First, the evaluation makes **fold-independent transformer initialization** explicit: pretrained ChemBERTa weights are restored before fine-tuning in every fold, removing the cross-fold weight-transfer channel that can inflate apparent transformer performance under cross-validation. This protocol detail is rarely reported in molecular benchmarks and is required for a valid transformer comparison. Second, the topological-fusion arms (GIN–TFP, GIN–TNE) test persistent-homology and tensor-network descriptors inside a graph architecture under scaffold-held-out evaluation on an antimalarial natural-product panel. The molecular benchmark is the primary analysis; a phenotype-only LISH reference is provided as orthogonal context in the Supporting Information (Section S1).

Under the scaffold split, **ECFP4–RF achieves the highest ROC AUC (0.830 0)**, ahead of GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7) and ChemBERTa (0.786 7); all four learned arms are significantly lower after multiplicity correction. Under the random split the ordering is unchanged (ECFP4–RF 0.943 3; all paired $p < 0.0001$ after Benjamini–Hochberg correction). Representation scale alone does not guarantee predictive gains on this panel. The topological fusion arm narrows the gap modestly, while projection-weight analysis identifies persistent-image dimensions as a salient block; this is presented as descriptive evidence about representation use, not causal attribution.

As an external molecule-disjoint transfer analysis, we repeated the ECFP4–RF versus GIN comparison on a molecule-disjoint ChEMBL-derived *P. falciparum* activity panel ($n = 22\,267$ after exclusion of 180 canonical-SMILES overlaps). ECFP4–RF again outperformed GIN under the random (0.954 7 versus 0.923 7) and scaffold (0.919 0 versus 0.884 3) splits. This extends the comparison beyond the internal screening labels, while the ChEMBL labels remain thresholded IC_{50}/EC_{50} records rather than prospective experiments. The benchmark supports a bounded conclusion: under the evaluated panels, split protocols, and model configurations, the tested learned representations did not improve on ECFP4–RF. We present this as a dataset- and protocol-bounded result, not as a statement about learned molecular models in general. Code, frozen splits and trained checkpoints are available in the public repository (`https://github.com/NanaEngo/Malaria_codesV2`).

This work is original, is not under consideration elsewhere, all authors have approved the submission, and the authors declare no competing financial interests.

We thank you for your consideration.

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